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AKDENİZ UNİVERSİTY • COMPUTER ENGİNEERİNG
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# FinTreX

## AI Agent Integration Planning Document

Homework 2 — Website Development & AI Agent Planning


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*"Bridging individual investors with professional financial consultants through a data-driven, AI-augmented consultation ecosystem."*

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# §1 1 Project Overview

## 1.1 Topic and Purpose

**FinTreX** is a comprehensive, multi-platform Financial Portfolio Management System designed to bridge the gap between individual investors and licensed financial professionals. The platform allows users to track, analyze, and manage their investment portfolios across three major asset classes: **BIST** (Borsa Istanbul), **Cryptocurrency**, and **Precious Metals**.

The core purpose of FinTreX is not simply to display market data — it is to create a **structured consultation ecosystem** where financial professionals (Economists) are integrated directly into the user's investment workflow. By combining real-time portfolio tracking with an assignment-based task system and encrypted in-app communication, FinTreX empowers users to make highly informed, data-driven financial decisions with the support of human expertise.

**Core Value Proposition**

FinTreX does not replace financial professionals. It makes them **more effective** by providing a centralized, data-rich environment and, in future phases, an AI layer that accelerates their analytical workflow.

## 1.2 Target Users

The system is designed around three distinct user roles, each with clearly bounded permissions and interface scopes:

**Table 1.** FinTreX User Roles and Characteristics

| Role | Description | Primary Platform | Access Level |
|---|---|---|---|
| **Standard User (Investor)** | Individual investors ranging from novices to experienced traders. Manages personal portfolios and interacts with assigned Economists. | Mobile (Flutter) & Web (Angular) | Standard |
| **Economist (Consultant)** | Licensed financial professionals. Provides analysis and fulfills analytical tasks. **Strictly read-only** access to user portfolios — cannot execute trades under any circumstances. | Web (Angular) — primary | Read-Only |
| **Administrator** | Platform operators managing users, Economist onboarding, subscription configuration, and system health. | Web (Angular) — exclusive | Full Admin |

## 1.3 Subscription Tiers

FinTreX operates on a tiered subscription model that governs feature access limits. This model is central to the platform's business logic and directly affects how the AI agent is surfaced to users.

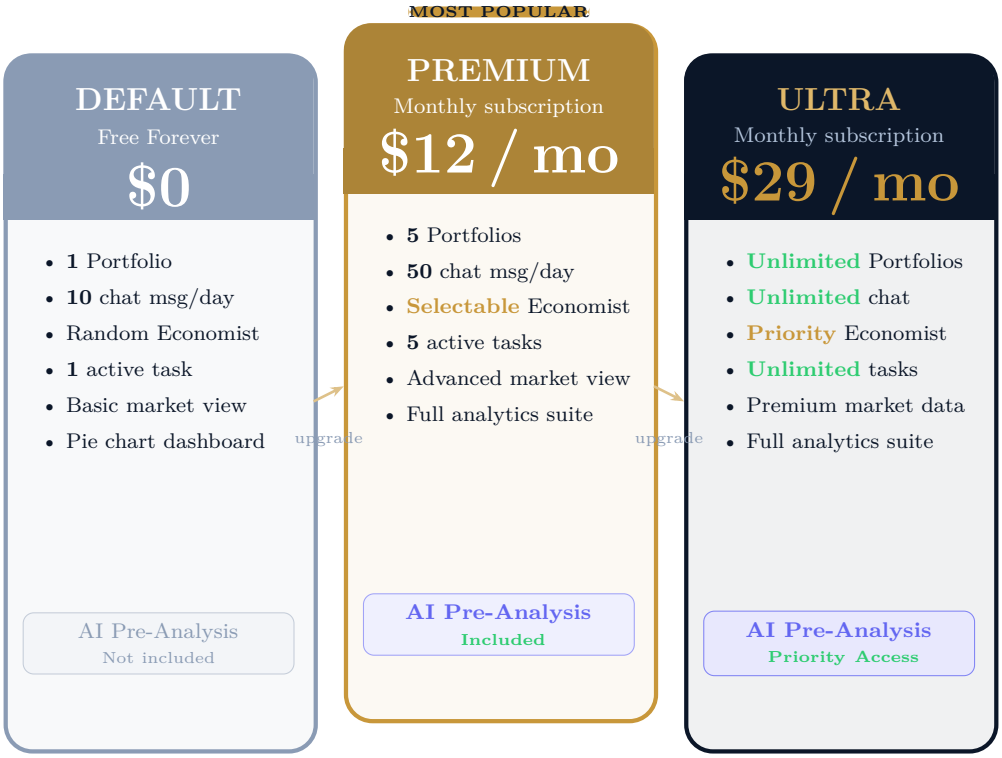



**Figure 1.** FinTreX Subscription Tiers — Feature and AI Access Comparison

## 1.4 Core Features

- **Multi-Market Portfolio Tracking:** Real-time asset tracking across BIST, Crypto, and Precious Metals via third-party financial APIs (Binance, BIST data providers).
- **Dynamic Visual Analytics:** Interactive pie charts and P&L (Profit & Loss) indicators rendered client-side using Chart.js (Angular equivalent: Apache ECharts).
- **Consultancy Ecosystem:** A structured task assignment system where users formally request market analyses from their assigned Economists, with tracked status (Pending → In Progress → Completed).
- **Built-in Encrypted Chat:** Subscription-limited real-time messaging between Users and Economists, with daily limits enforced server-side.
- **Tiered Subscription Management:** Integrated Stripe payment processing for Premium and Ultra plan lifecycle management.
- **Administrative Panel:** Full-featured management interface for account auditing, Economist onboarding verification, and live subscription configuration.

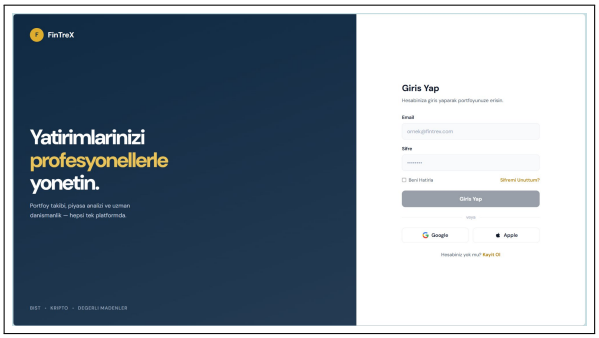
## 1.5 Planned Technology Stack

**Table 2.** FinTreX Production Technology Stack

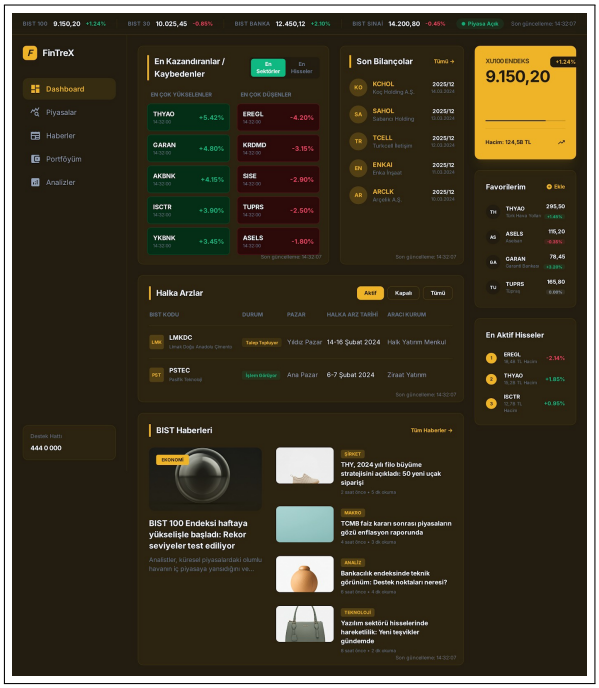
| Layer | Technology | Purpose |
|---|---|---|
| **Web Frontend** | Angular 17+ (TypeScript) | SPA for all three user roles |
| **Mobile App** | Flutter (Dart) | Cross-platform iOS & Android |
| **Backend API** | .NET 8 (C#), REST | Business logic & endpoints |
| **Database** | Azure SQL + EF Core 8.x | Relational data store |
| **Authentication** | JWT (JSON Web Tokens) | Stateless, role-based auth |
| **Payments** | Stripe API | Subscription transactions |
| **Market Data** | Binance API, BIST Providers | Live asset pricing |
| **Cloud / DevOps** | Microsoft Azure + Azure DevOps | Hosting, CI/CD, Key Vault |

## 1.6 Draft UI — Frontend Prototype Screenshots

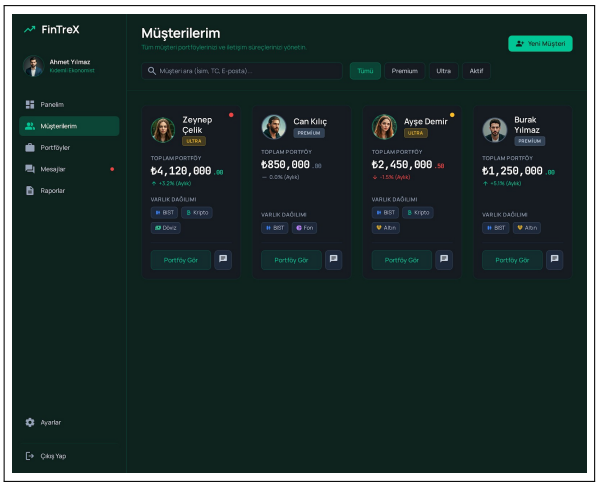
All UI designs were created using **Stitch**, an AI-powered UI generation tool, and then implemented as a functional HTML/CSS/JS prototype. The Stitch mockups cover all major modules of the system across three user roles: Standard User, Economist, and Administrator.



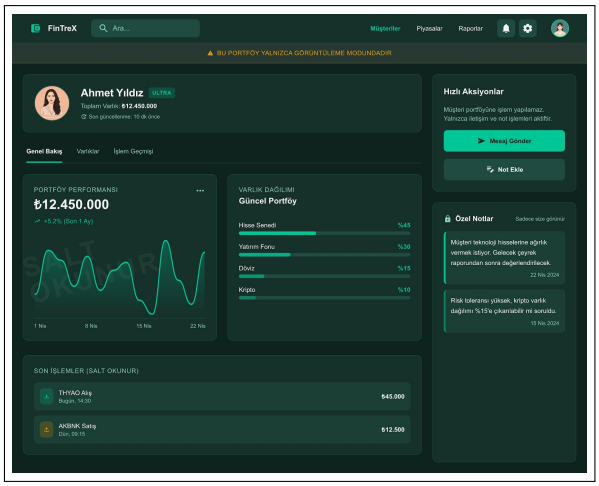

**(a) Login Page** — Split-layout authentication screen. Left panel shows the FinTreX brand and value proposition; right panel contains the login form with Google/Apple SSO options and a "Kayıt Ol" registration link. JWT is issued on success and the user is routed by role.



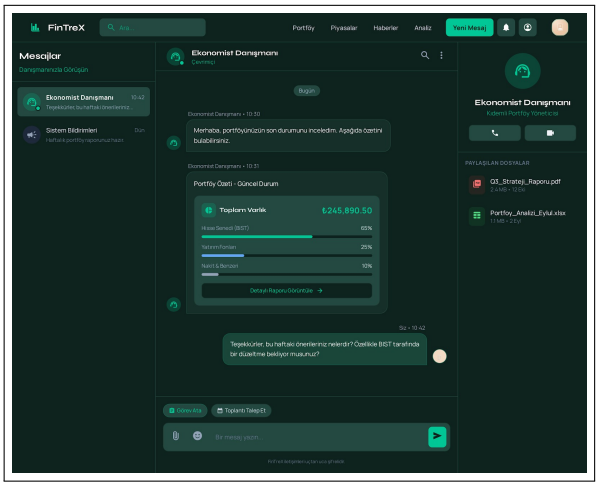

**(b) Standard User — BIST Dashboard** — Real-time market ticker at top (BIST 100, BIST 30, BIST Banka). Shows top gainers/losers, recent balance reports, IPO listings, BIST news feed, and a Favorites watchlist. XU100 index card visible top-right with live volume data.



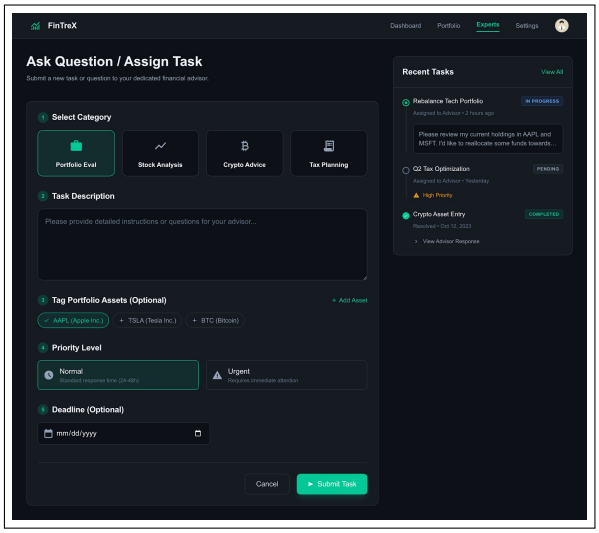

**(a) Economist — Müşterilerim (My Clients)** — Client management panel for the Economist role. Displays client cards with subscription tier badges (Ultra/Premium), total portfolio values, monthly P&L, and asset class distribution tags (BIST, Kripto, Altın). Filter tabs allow sorting by tier.



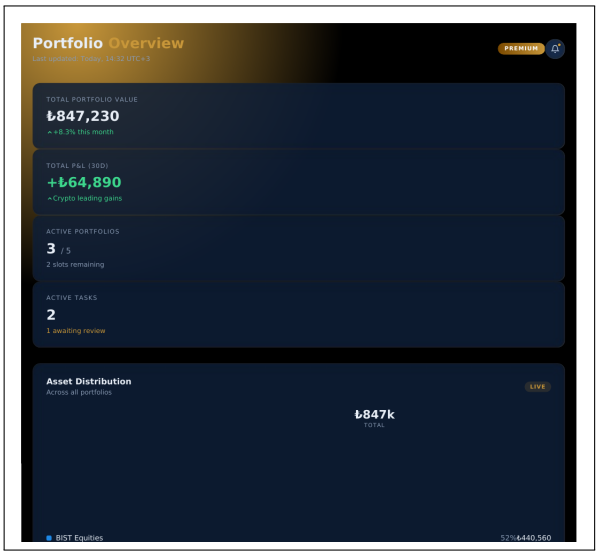

**(b) Economist — Read-Only Portfolio View** — The yellow banner "BU PORTFÖY YALNIZCA GÖRÜNTÜLEME MODUNDADIR" explicitly enforces the read-only constraint. The Economist sees performance charts, asset distribution, and recent transactions — but the only available actions are "Mesaj Gönder" and "Not Ekle". No buy/sell or trade execution buttons exist on this screen.



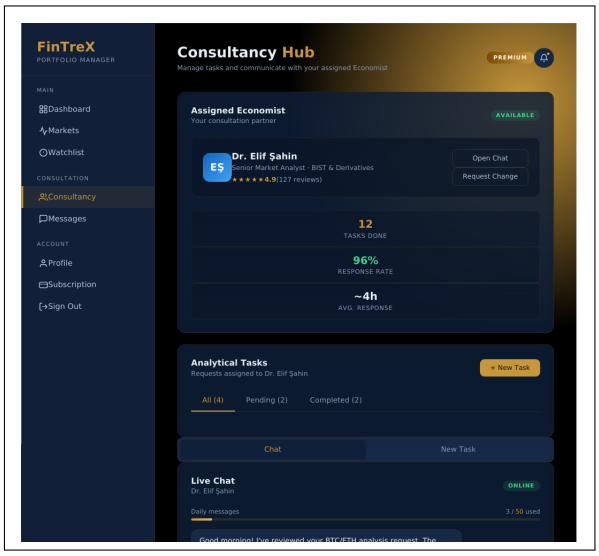

(a) **User — Mesajlar (Chat with Economist)** — Encrypted in-app chat interface. The Economist can send structured portfolio summary cards directly within the chat thread. Shared files panel on the right. "Görev Ata" and "Toplantı Talep Et" quick-action buttons at the bottom.




(b) **User — Task Assignment Form (AI Trigger Point)** — Five-step task submission form: category selection (Portfolio Eval, Stock Analysis, Crypto Advice, Tax Planning), task description, asset tagging, priority level, and deadline. This is the exact moment the **PAA AI agent is triggered** in the planned architecture — upon "Submit Task", the AI pipeline activates automatically.




(a) **Prototype — Dashboard** — Functional HTML/CSS/JS implementation of the dashboard. KPI cards (847k total, +8.3% P&L), Chart.js doughnut chart, Holdings table, and Market Snapshot ticker — all driven by mocked JSON data.




(b) **Prototype — Consultancy Hub** — Functional implementation of the task and chat module. Economist card, task status filtering, and Live Chat with daily message limit bar (Premium: 50 msg/day). No trade buttons present — read-only constraint enforced.

**Design & Implementation Note**

All UI screens were designed in **Stitch** (AI UI generation tool) and subsequently implemented as a functional **HTML/CSS/JS prototype**. The production migration to **Angular 17+** will preserve the established color language (Deep Navy · Slate Gray · Accent Gold / Teal) and component architecture. The **Task Assignment form** (Figure 9) is the designated AI trigger point — submitting a task activates the PAA pipeline described in Section 2.

# §2 1 AI Agent Concept

## 2.1 Agent Selection: Pre-Analysis Assistant (PAA)

Two AI agent concepts were evaluated for FinTreX:

**Table 3.** AI Agent Candidate Evaluation

| Candidate | Description / Trade-offs | Decision |
|---|---|---|
| **AI Terminology Assistant** | A chatbot for Standard Users to ask basic financial questions (e.g., "What is P&L?"). Generic feature; does not leverage FinTreX's unique data model. | Deferred |
| **Pre-Analysis Assistant (PAA)** | Automatically generates a portfolio summary for the Economist when a task is submitted. Solves a unique, high-value bottleneck in the FinTreX consultation workflow. Architecturally consistent with the read-only constraint. | Selected |

**Rationale for PAA:** The Terminology Assistant is a useful but generic feature widely available through standard LLM integrations. The PAA, by contrast, addresses a specific bottleneck *unique to the FinTreX model*: the "cold-start" problem that Economists face when a task arrives and they must parse a client's full portfolio from scratch before they can begin analysis.

## 2.2 Problem the Agent Solves

In the current (non-AI) FinTreX workflow, when a Standard User assigns a task — for example, *"Analyze the risk exposure of my BIST allocation"* — the Economist must:

1. Manually open the user's read-only portfolio view.
2. Parse the current asset distribution, quantities, and P&L status.
3. Cross-reference current market conditions from external sources.
4. Form a preliminary understanding of the portfolio context before drafting their response.

This **cold-start process** consumes valuable Economist time, especially for professionals managing multiple clients across different subscription tiers. The Pre-Analysis Assistant automates steps 1–3 by delivering a structured, data-driven summary report directly to the Economist's task interface *before* they begin their review.

**Important Scope Constraint**

The PAA does **not** make financial recommendations, generate buy/sell signals, or provide investment advice. Its sole function is to summarize *already available, objective portfolio and market data* in a structured format. Financial advice originates *exclusively* from licensed human Economists — this is both a platform design principle and a legal constraint.

## 2.3 Agent Type and Classification

**Table 4.** PAA Agent Classification

| Property | Description |
|---|---|
| **Agent Type** | Goal-based, Context-Aware Summarization Agent |
| **Trigger Mechanism** | Event-driven: fires when a new analytical task is created |
| **Input Sources** | Portfolio snapshot (Azure SQL) + cached market data + user's task description |
| **Output Artifact** | Structured pre-analysis report (JSON/text) stored in Azure SQL |
| **LLM Provider** | External API — Anthropic Claude API or OpenAI GPT-4o |
| **Autonomy Level** | Non-autonomous — read-only, no decision-making, no action execution |
| **Data Access** | Read-only to portfolio and market tables only |

## 2.4 User Interaction and Workflow

The PAA operates **silently in the background** and surfaces its output at a single, well-defined moment in the workflow. The interaction flow is described below:

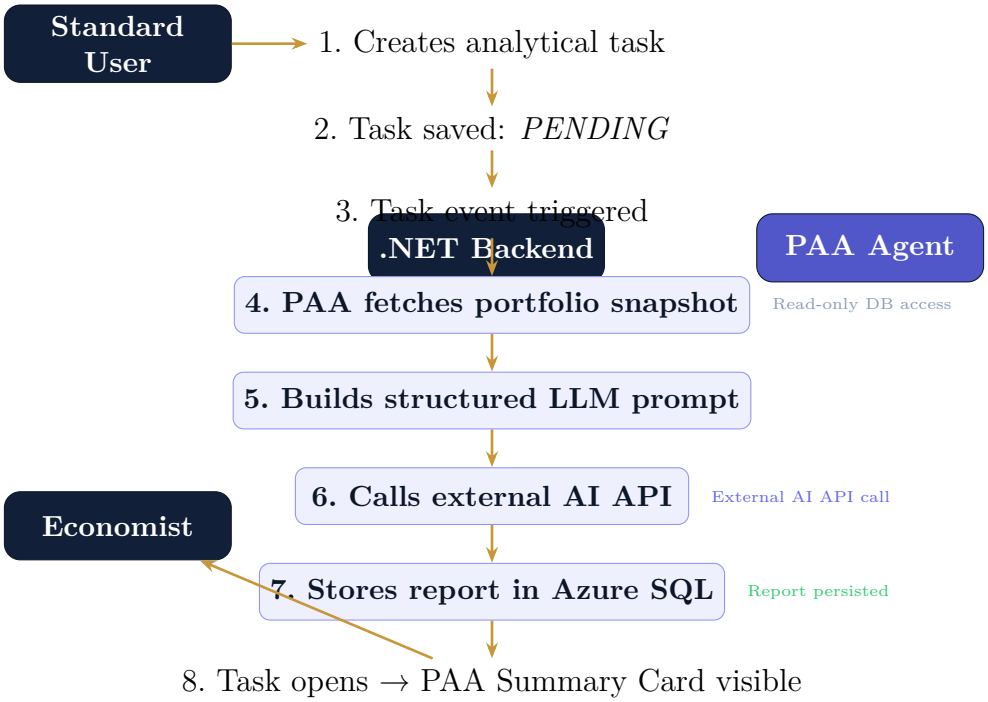



**Figure 6.** PAA Agent Trigger and Execution Workflow

### PAA Summary Card — Mockup Content

When the Economist opens the task, they see a clearly labelled AI Summary Card *above* the standard portfolio view. An example of the card's content is shown below:

**Pre-Analysis Summary Card (AI-Generated · For Context Only)**

**Portfolio Snapshot:** User holds 3 active portfolios. BIST allocation: 52% (dominant: THYAO, AKBNK). Cryptocurrency: 31% (dominant: BTC, ETH). Precious Metals: 17% (Gold XAU/USD).

**P&L Status:** Total portfolio is **+8.3%** over the last 30 days. Crypto holdings show the highest volatility (±12% weekly). BIST holdings are +4.1% MTD.

**Task Context:** User's task references BIST allocation review. BIST100 index has declined 2.1% over the past 7 days.

*Disclaimer: This summary is AI-generated from objective portfolio data. It does not constitute financial advice. Analysis and recommendations are the exclusive responsibility of the assigned Economist.*

# §3 1 System Architecture

## 3.1 High-Level Overview

FinTreX follows a **client-server architecture** with three distinct layers: the frontend client tier, the centralized .NET backend tier, and the external services tier. The AI Agent (PAA) is integrated as an internal backend microservice, triggered by task events

and communicating outbound to an external LLM API.

## 3.2 Architecture Diagram

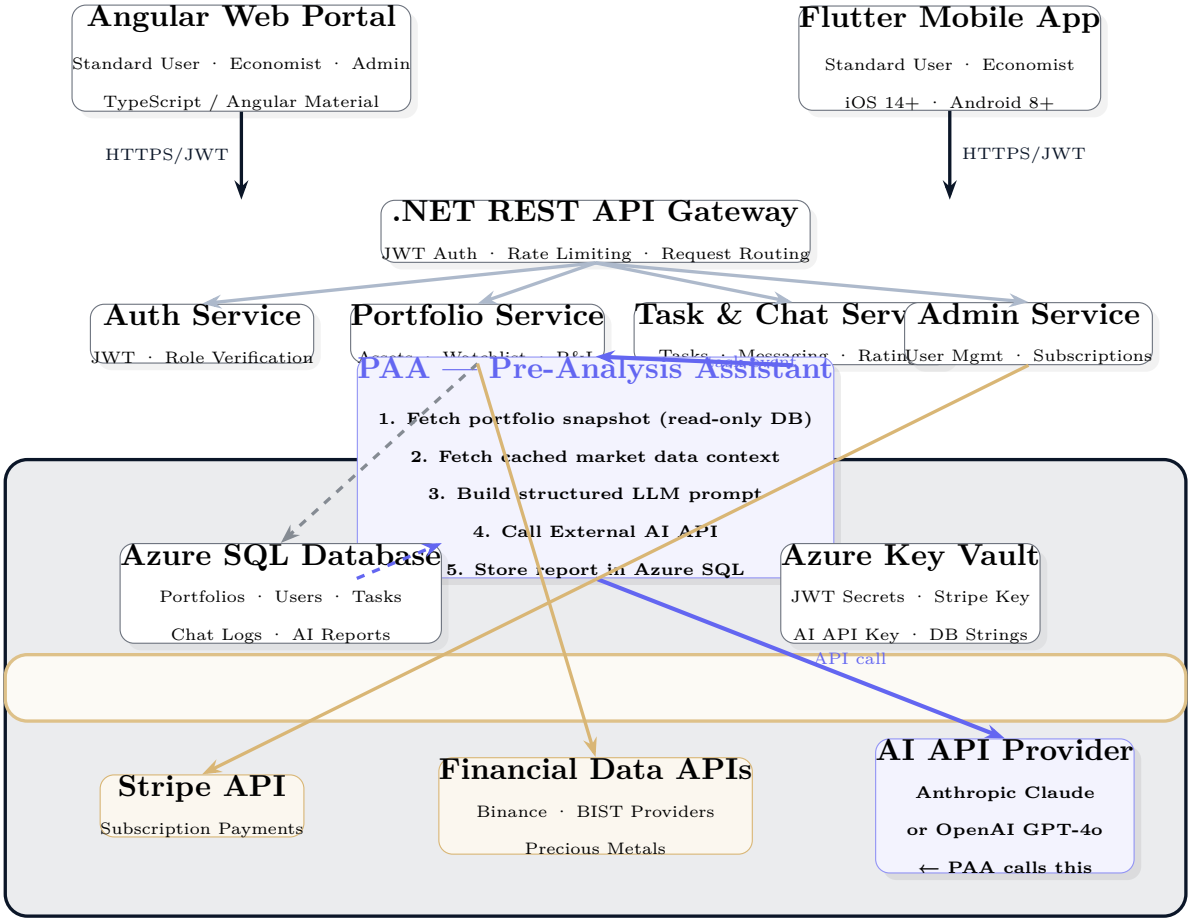



**Figure 7.** FinTreX System Architecture — including PAA Agent Integration

## 3.3 PAA Data Flow — Step by Step

The following table details the exact data flow when the AI agent pipeline is triggered:

**Table 5.** PAA Agent Data Flow

| # | Step | Detail |
|---|---|---|
| 1 | **Task Created** | User submits a task via Angular/Flutter. `POST /api/tasks` received by .NET Gateway. |
| 2 | **Task Persisted** | Task Service saves task to Azure SQL with status `PENDING`. An internal task-created event is emitted. |
| 3 | **PAA Initialized** | PAA service subscribes to task events. Retrieves user's full portfolio snapshot and latest cached market data from Azure SQL (**read-only**). |
| 4 | **Prompt Built** | Structured prompt constructed: portfolio data (JSON), task description (plain text), and a system instruction constraining the LLM to produce only objective, factual summaries — no advice. |
| 5 | **AI API Called** | `POST` to external AI API endpoint. API key retrieved securely from Azure Key Vault. |
| 6 | **Response Processed** | LLM response received, validated for format, and stored as a `pre_analysis_report` record in Azure SQL, linked to the task. |
| 7 | **Delivered to Economist** | When Economist opens the task, `GET /api/tasks/{id}` returns task + PAA report. Frontend renders the AI Summary Card above the portfolio view. |

### 3.4 Security and Compliance

**Table 6.** PAA Security and Compliance Measures

| Concern | Measure |
|---|---|
| **Portfolio Safety** | PAA service has database permissions restricted to **read-only** on portfolio and market tables. It cannot insert, update, or delete user assets under any circumstances. |
| **User Data Privacy** | User PII (name, email, national ID) is **stripped** from the payload before calling the external AI API. Only aggregated asset types, quantities, and values are transmitted. |
| **Transparency** | All AI-generated content is clearly labelled with a disclaimer. Economists are always aware they are reviewing an AI summary, not a human analysis. |
| **API Key Security** | AI provider API key stored exclusively in **Azure Key Vault**. Never embedded in source code, environment files, or client-side assets. |
| **Advice Scope Limitation** | LLM system prompt explicitly prohibits buy/sell recommendations, price predictions, or investment advice. Violations are filtered before the report is stored. |

### 3.5 Future Enhancements (Post-MVP Roadmap)

**Table 7.** Planned AI Feature Roadmap

| Feature | Phase | Description |
|---|---|---|
| **Watchlist Anomaly Alerts** | Phase 2 | PAA monitors user watchlists and sends proactive push notifications when abnormal price movements are detected. |
| **AI Terminology Assistant** | Phase 2 | Secondary LLM chatbot for Standard Users to query basic financial terminology (the deferred candidate agent). |
| **Economist Productivity Dashboard** | Phase 3 | Aggregated AI insights for Economists showing behavioral trends and patterns across their full client portfolio book. |
| **Sentiment Analysis Module** | Phase 3 | Integration with financial news APIs to enrich PAA reports with real-time market sentiment context. |